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Nucleotide sequence and distribution of gene *tetO* encoding tetracycline resistance in *Campylobacter coli.*

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1. SUMMARY

The nucleotide sequence of the tetracycline resistance gene encoded by the self-transferable plasmid pIP1433 in *Campylobacter coli* BM2509 has been determined. The resistance gene, *tetO,* was located in an open reading frame of 1917 bp corresponding to a protein with a calculated M_r of 72 290. The transcriptional and translational signals have been identified. The *tetO* gene exhibits 76% sequence identity with *tetM* from *Streptococcus* and the deduced amino acid sequences are 77% homologous. The substitutions are scattered throughout indicating that the two genes have diverged from a common ancestor. We suggest that tetracycline resistance in *Campylobacter* is due to acquisition of a gene originating in a Gram-positive bacterium. The distribution of *tetO* in hundred strains of *C. jejuni* and *C. coli* resistant to tetracycline (MIC $\geq$ 32 μg/ml) and to minocycline (MIC $\geq$ 4 μg/ml) was studied by dot blot hybridization under high stringency conditions using a 1.4-kb intragenic probe. Homology with *tetO* was detected in 98 strains isolated from humans and animals in various geographical areas. None of the strains hybridized with *tetM* under the same conditions. These data indicate that *tetO* is widespread among *Campylobacter* spp. probably due to conjugative plasmids.

2. INTRODUCTION

Tetracycline resistance (Tc^R) is widespread among numerous Gram-positive and Gram-negative bacterial genera. In Gram-negative bacteria, a minimum of five classes of Tc^R genes (*tetA* to *tetE*) have been described on the basis of nucleic acid hybridizations [1,2]. Resistance in bacteria harbouring these genes is achieved by active efflux of tetracycline from the cells [3]. In streptococci, three genetically distinct Tc^R determinants (*tetL, tetM* and *tetN*) have been identified which are not structurally related to the Tc^R genes of Gram-negative bacteria [4]. The *tetL* determinant probably specifies active efflux of tetracycline whereas *tetN* and *tetM* mediate Tc^R at the level of protein synthesis [3].

Campylobacter coli and *C. jejuni* are Gram-negative bacteria frequently responsible for

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bacterial acute gastroenteritis in humans. Tetracycline resistance in *Campylobacter* is generally encoded by plasmids which are able to transfer to other *Campylobacter* spp. but not to *Escherichia coli* [5,6]. No homology was found between the Tc^R determinants of *Campylobacter* and those of Gram-negative bacteria [5]. On the basis of DNA annealing studies, it was reported that the plasmid-borne Tc^R determinant in *Campylobacter* was homologous to the streptococcal *tetM* gene [7]. Martin et al. reached an opposite conclusion by similar experiments under stringent conditions [8]. Hybridization experiments performed under various stringency conditions lead us to conclude that the Tc^R determinant of *Campylobacter* was structurally related to but distinct from *tetM*. We therefore determined the nucleotide sequence of the Tc^R gene, designated *tetO*, carried by the self-transmissible plasmid pIP1433 in *C. coli* BM2509. A probe specific for this gene was used to study the distribution of this R determinant in the genus *Campylobacter*.

3. MATERIALS AND METHODS

3.1. Bacterial strains, plasmids and phages

One hundred Tc^R strains of *C. coli* and *C. jejuni* isolated from different geographical areas: France, 51; Vietnam, 15; Thailand, 13; Switzerland, 10; and Algeria, 8; were studied. Eighty-two were clinical isolates and the remaining was from porcine origin. *C. coli* BM2509 harbouring plasmid pIP1433 (Km^R Tc^R, 47.2 kb) was described previously [9]. Strains *C. jejuni* MK175 [5] and *C. coli* 981 [10] were included. *E. coli* JM103 [11] was transfected with DNA fragments to be sequenced, which have been cloned in bacteriophage M13mp18 or M13mp19 [12]. Plasmid pUC19 (Ap^R, 2.7 kb) [13] was used in cloning experiments. Recombinant plasmids were introduced by transformation [14] into *E. coli* HB101 [15].

3.2. Media and growth conditions

Campylobacter strains were grown on Mueller-Hinton agar supplemented with 5% defibrinated horse blood and containing (μg/ml) vancomycin, 10; trimethoprim, 5; polymyxin B, 30; and cephalotin, 15. All incubations were at 37°C for 48 h in a microaerophilic atmosphere (5% O_2, 10% CO_2) provided by a gas generating kit (Oxoid).

3.3. Determination of minimal inhibitory concentrations

The method of Steers et al. [16] was used to determine the MICs of tetracycline and minocycline.

3.4. Mating experiments

C. fetus BM2560 (Nal^R Rif^R) [9] was used as a recipient. Donor and recipient strains were mixed in the same ratio and mating was performed on a nitrocellulose filter for 48 h. Transconjugants were selected on blood agar containing (μg/ml) nalidixic acid, 50 and tetracycline, 10.

3.5. Preparation of plasmid DNA

Plasmid DNA was purified as described [14].

3.6. DNA–DNA hybridization

The purified restriction DNA fragments used as probes (Table 1) were labelled in vitro with [α-^{32}P]-dATP by nick translation [14]. Dot blot hybridization under stringent conditions was in 50% formamide at 42°C for 24 h followed by two washings in 0.2 × SSC, 0.1% SDS at 65°C for one hour. Hybridization under low stringency conditions was in 50% formamide at 37°C for 24 h and

Table 1

DNA probes constructed and used in hybridization experiments.

Gene	Probe	Reference or origin
tetK	870-bp *Hin*cII fragment of pT181	26
tetL	310-bp *Cla*I-*Hpa*II fragment of pBC16	27, 28
tetM	850-bp *Cla*I-*Hin*dIII fragment of Tn1545	8
tetO	1458-bp *Hin*dIII-*Nde*I fragment of pIP1433	this paper
tetN	45-kb plasmid pMV120	29

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washing was in 2 × SSC, 0.1% SDS at 45 °C for 1 h.

3.7. DNA sequencing

DNA fragments cloned into bacteriophage M13 derivatives were treated with T4 DNA polymerase, as described by Dale et al. [17], to generate sequential series of overlapping clones which were sequenced by the dideoxynucleotide chain terminator technique [18].

3.8. Computer analysis of sequence data

The DNA sequences were arranged using DBCOMP and DBUTIL computer programs [19]. DNA and protein sequences were compared using the algorithm of Wilbur and Lipman [20].

3.9. Enzymes and biochemicals

Restriction endonucleases, calf intestinal phosphatase, DNA polymerase I (Klenow fragment), T4 DNA ligase and radioactive chemicals were purchased from Amersham International. Deoxynucleoside triphosphates, dideoxynucleoside triphosphates, and M13 pentadecamer primer were from Biolabs. Bacteriophages M13mp18 and M13mp19 RF DNA were from PL-Biochemicals. The antibiotics were provided by the following laboratories: kanamycin, Bristol; trimethoprim, Roche; tetracycline and minocycline, Pfizer; vancomycin and cephalotin, Eli Lilly; polymyxin B, Roger Bellon; nalidixic acid, Winthrop; ampicillin, Rhone-Poulenc.

4. RESULTS AND DISCUSSION

4.1. Cloning and localisation of the Tc^R gene of pIP1433

Plasmid pIP1433 and pUC19 DNA digested with *Bgl*II and *Bam*HI, respectively were mixed, ligated and introduced by transformation into *E. coli* HB101. Transformants resistant to ampicillin and tetracycline were screened for their plasmid content by agarose gel electrophoresis. The smallest hybrid plasmid, pAT121, was found to contain a 5-kb pIP1433 *Bgl*II DNA insert (Fig. 1). Subcloning was accomplished by introduction of the 4-kb *Eco*RI fragment of the insert of pAT121 into the *Eco*RI site of pUC19, generating plasmids pAT122 and pAT123, containing the Tc^R region in either orientation (data not shown). All attempts to obtain Tc^R clones containing the 1.6-kb or the 1.8-kb *Hin*dIII DNA fragment (Fig. 1) were unsuccessful suggesting that these two fragments are necessary for expression of Tc^R. The restriction endonuclease map of the insert of pAT121 (Fig. 1) was found to differ from that of the 2.9-kb *Hin*cII-BAL-31-generated DNA fragment containing the *tetM* gene of the streptococcal transposon Tn1545 [8].

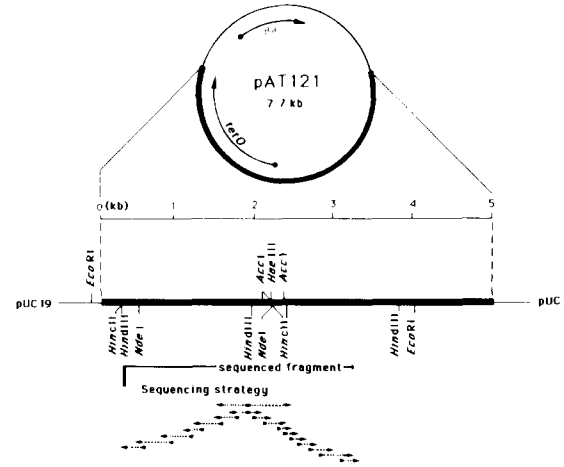

Fig. 1. Restriction map of the 5-kb insert in plasmid pAT121 and sequencing strategy for determination of the nucleotide sequence of *tetO*. Top: plasmid pAT121 was constructed by inserting a 5-kb *Bgl*II fragment of pIP1433 (heavy black line) into the *Bam*HI site of pUC19 (thin line). Curved arrows indicate direction and extent of transcription of resistance genes. Bottom: partial restriction map and sequencing strategy of the DNA region containing *tetO*. Insert and vector segments are depicted by heavy black line and thin line, respectively. Dashed arrows indicate the direction and extent of the dideoxy-sequencing reactions.

4.2. Nucleotide sequence of the tetO gene

The sequencing strategy for the *tetO* gene is depicted in Fig. 1. A 2.7-kb DNA fragment including the Tc^R determinant of pIP1433 was se-

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       CATAAACAGA TGATTAGTGG CAGGGGGAAA TCCTGCCGCT TTTTCTGCTT TAGTTTGTCA GCTTGACAAA TAAAGGGTTA AGGAATATAA TTAGATTCAG TATTAT  464

      ACAA GGAGTTAATA AATATGCGGC AAGGTATTCT TAAATAAACT GTCAATTTGA TAGTGGGAAC AAAAAGTAGC AGTCGCCGTT TCACTTTTAA TATGGGGCTT AGTT  572

      TTTTGT ACCCAGTTTA AGAATACTTT TATCATGTAA TTTTATATGC CCGAAAACAT ATAAGTGTTT TGGGGCTATT GGAGTTATTT A--------C CCAGTGATAG GA  672
                                   G+ ++C++A+GTA T+CGG+CA++ ++CT++A+GC +TT+TA+GCC TATGGTTA+G CATAAAAAT+ ++++++++A ++   74

                                                  RBS                  MET LYS ILE ILE ASN LEU GLY ILE LEU ALA HIS VAL ASP ALA
TETO
tetO  GTATTTAT CACTGGGTAT TTTTATGCCC TTTTTTGGGT GTTGATAGGA GCAAAATTAC ATG AAA ATA ATT AAC TTA GGC ATT CTG GCT CAC GTT GAC GCA  772
tetM  ++++++++ +++++++-++ ++++++++++ ++++GG+CT+ -+++AT+++ ++++++C++ +++ +++ ++T +++ +++ A+T ++A G++ T+A +++ ++T +++ ++T ++G  172

TETO  GLY LYS THR THR LEU THR GLU SER LEU LEU TYR THR SER GLY ALA ILE ALA GLU PRO GLY SER VAL ASP LYS GLY THR THR ARG THR ASP
tetO  GGA AAG ACA ACA TTA ACG GAA AGT TTA TTG TAT ACC AGT GGT GCA ATT GCA GAA CCA GGG AGC GTA GAT AAA GGC ACA ACA AGG ACA GAT  862
tetM  +++ ++A ++T ++C +++ ++A +++ ++C +++ ++A +++ +A+ +++ ++A ++G +++ A++ +++ TT+ ++A +++ ++C ++C +G+ ++T +++ ++G +AA ++G +++  262

TETO  THR ILE ASN LEU GLU ARG GLN ARG GLY ILE THR ILE GLN THR ALA VAL THR SER PHE GLN TRP GLU ASP VAL LYS VAL ASN ILE ILE ASP
tetO  ACA ATT AAT TTG GAG CGT CAA AGG GGA ATC ACT ATC CAG ACA GCA GTG ACA TCT TTT CAG TGG GAG GAT GTA AAA GTC AAC ATT ATA GAT  952
tetM  +AT +CG CT+ ++A ++A +++ ++G ++A +++ ++T ++A ++T +++ ++G ++G A+A ++C +++ +++ +++ +++ A+A A++ ACT ++G ++G +++ ++C +++ ++C  352

TETO  THR PRO GLY HIS MET ASP PHE LEU ALA GLU VAL TYR ARG SER LEU SER VAL LEU ASP GLY ALA VAL LEU LEU VAL SER ALA LYS ASP GLY
tetO  ACG CCA GGC CAT ATG GAT TTT TTG GCG GAA GTA TAC CGT TCT TTA TCC GTA TTA GAC GGA GCA GTA TTA TTA GTT TCT GCA AAG GAT GGC 1042
tetM  +++ +++ ++A +++ +++ +++ +++ ++A ++A +++ +++ ++T +++ ++A +++ ++A +++ +++ ++T ++G +++ A+T C++ C+G A++ +++ +++ ++A +++ +++  442

TETO  ILE GLN ALA GLN THR ARG ILE LEU PHE HIS ALA LEU GLN ILE MET LYS ILE PRO THR ILE PHE PHE ILE ASN LYS ILE ASP GLN GLU GLY
tetO  ATA CAG GCA CAG ACC CGT ATA CTG TTT CAT GCA CTA CAG ATA ATG AAG ATT CCG ACA ATT TTT TTC ATC AAT AAA ATT GAC CAA GAG GGG 1132
tetM  G++ ++A +++ ++A ++T +++ +++ T++ +++ +++ +++ ++T AG+ +A+ ++A GCT +++ ++C +++ ++C +++ +++ ++G +++ +++ +++ A+T ++A  532

TETO  ILE ASP LEU PRO MET VAL TYR GLN GLU MET LYS ALA LYS LEU SER SER GLU ILE ILE VAL LYS GLN LYS VAL GLY GLN HIS PRO HIS ILE
tetO  ATT GAT TTG CCA ATG GTA TAT CAA GAA ATG AAA GCA AAG CTT TCT TCG GAA ATT ATA GTG AAG CAA AAG GTT GGG CAG CAT CCC CAT ATA 1222
tetM  +++ +++ ++A T++ +C+ ++T +++ ++G ++T ++T +++ +AG ++A +++ +++ G++ +++ +++ G++ A+C ++A ++G +++ ++A +AA +T+ +++ ++T A++ ++G  622

TETO  ASN VAL THR ASP ASN ASP ASP MET GLU GLN TRP ASP ALA VAL ILE MET GLY ASN ASP GLU LEU LEU GLU LYS TYR MET SER GLY LYS PRO
tetO  AAT GTA ACG GAC AAT GAC GAT ATG GAA CAG TGG GAT GCG GTA ATT ATG GGA AAC GAT GAA CTA TTA GAG AAA TAT ATG TCA GGG AAA CCG 1312
tetM  CG+ +++ +T+ A++ TT+ AC+ ++A TCT +++ ++A +++ +++ AT+ +++ ++A GAA +++ ++T +++ T+C ++T ++G +++ +++ +++ +C+ ++T +++ +++ TTA  712

TETO  PHE LYS MET SER GLU LEU GLU GLN GLU GLU ASN ARG ARG PHE GLN ASN GLY THR LEU PHE PRO VAL TYR HIS GLY SER ALA LYS ASN ASN
tetO  TTT AAA ATG TCA GAA CTG GAA CAG GAA GAA AAC AGG AGA TTC CAA AAC GGA ACG TTA TTT CCC GTT TAT CAC GGA AGC GCT AAA AAC AAT 1402
tetM  ++G G++ GCA +T+ +++ ++C +++ ++A ++G +++ +G+ +TA +++ ++T ++T ++T T+T T+C C+G ++C ++T +++ +++ +++ +++ ++T ++A +++ +++ +++  802

TETO  LEU GLY ILE ARG GLN LEU ILE GLU VAL ILE ALA SER LYS PHE TYR SER SER THR PRO GLU GLY GLN SER GLU LEU CYS GLY GLN VAL PHE
tetO  CTG GGG ATT CGG CAG CTT ATA GAA GTA ATT GCC AGT AAA TTT TAT TCA TCA ACG CCT GAA GGT CAA TCT GAA CTA TGC GGG CAG GTT TTT 1492
tetM  A+A +++ +++ GAT A+C +++ +++ +++ ++G +++ A+G +A+ +++ +++ +++ +++ +++ ++A +A+ CG+ +++ ++G +++ +++ ++T +++ ++A A+A +++ ++C  892

TETO  LYS ILE GLU TYR SER GLU LYS ARG ARG ARG PHE VAL TYR VAL ARG ILE TYR SER GLY THR LEU HIS LEU ARG ASP VAL ILE ARG ILE SER
tetO  AAG ATT GAA TAT TCA GAG AAA AGG CGG CGT TTT GTT TAT GTG CGT ATA TAT AGC GGA ACA TTG CAT TTG AGG GAT GTT ATT AGA ATA TCT 1582
tetM  ++A +++ ++G +++ ++G +++ ++A +++ ++A +A+ +++ C++ +CA +++ A+A +++ C+T +++ ++T +++ C++ GT+ C++ +++ +++ C+A +++ CCG G++ +++ +++ ++G  982

TETO  GLU LYS GLU LYS ILE LYS ILE THR GLU MET CYS VAL PRO THR ASN GLY GLU LEU TYR SER SER ASP THR ALA CYS SER GLY ASP ILE VAL
tetO  GAA AAA GAG AAA ATA AAA ATC ACA GAG ATG TGT GTT CCG ACA AAC GGT GAA TTA TAT TCA TCC GAT ACA GCC TGC TCT GGT GAT ATT GTA 1672
tetM  +++ ++G ++A +++ +++ +++ ++T +++ ++A +++ +A+ AC+ T+A +T+ ++T +++ +++ +++ +C+ AA+ AT+ +++ +AG ++T +AT ++C ++G ++A +++ ++T 1072

TETO  ILE LEU PRO ASN ASP VAL LEU GLN LEU ASN SER ILE LEU GLY ASN GLU ILE LEU LEU PRO GLN ARG LYS PHE ILE GLU ASN PRO LEU PRO
tetO  ATT TTA CCA AAT GAT GTT TTG CAG CTA AAC AGT ATT TTG GGG AAC GAA ATA CTG TTG CCG CAG AGA AAA TTT ATT GAA AAT CCT CTC CCT 1762
tetM  +++ ++G +AG +++ ++G T++ +++ A++ T++ ++T +++ G++ C+T ++A G+T AC+ +AG ++A +++ ++A +++ +++ G+G AGA +++ +++ +++ ++C +++ +++ 1162

TETO  MET LEU GLN THR THR ILE ALA VAL LYS LYS SER GLU GLN ARG GLU ILE LEU LEU GLY ALA LEU THR GLU ILE SER ASP GLY ASP PRO LEU
tetO  ATG CTC CAA ACA ACG ATT GCA GTA AAG AAA TCT GAA CAG CGG GAA ATA TTG CTT GGG GCA CTT ACA GAA ATT TCA GAT GGC GAC CCT CTT 1852
tetM  C++ ++G +++ ++G ++T G++ +A+ CCG +GC +++ C++ C++ ++A A++ +++ ++G ++A +++ +AT +++ +++ TT+ +++ ++C ++C ++C A+T +++ ++G +++ 1252

TETO  LEU LYS TYR TYR VAL ASP THR THR THR HIS GLU ILE ILE LEU SER PHE LEU GLY ASN VAL GLN MET GLU VAL ILE CYS ALA ILE LEU GLU
tetO  TTA AAA TAT TAT GTG GAT ACT ACA ACG CAT GAG ATT ATA CTT TCT TTT TTG GGG AAT GTG CAG ATG GAA GTC ATT TGT GCC ATC CTT GAG 1942
tetM  C+G CG+ +++ +++ +++ +++ T++ G+G ++A +++ ++A ++C +++ +++ +++ ++C ++A +++ ++A ++A ++A +++ +++ ++G +C+ +++ ++T C+G ++G C+A 1342

TETO  GLU LYS TYR HIS VAL GLU ALA GLU ILE LYS GLU PRO THR VAL ILE TYR MET GLU ARG PRO LEU ARG LYS ALA GLU TYR THR ILE HIS ILE
tetO  GAA AAA TAT CAT GTG GAG GCA GAA ATA AAA GAG CCT ACT GTT ATA TAT ATG GAA AGA CCG CTT AGA AAA GCA GAA TAT ACC ATC CAC ATA 2032
tetM  +++ ++G +++ +++ +++ +++ AT+ +++ +++ +++ +++ +++ ++A ++C ++T +++ +++ +++ +++ +++ T+A +A+ +++ +++ ++G +++ +++ ++T +++ ++C 1432

TETO  GLU VAL PRO PRO ASN PRO PHE TRP ALA SER VAL GLY LEU SER ILE GLU PRO LEU LEU LEU GLY SER GLY VAL GLN TYR GLU SER ARG VAL
tetO  GAA GTC CCG CCA AAT CCT TTC TGG GCT TCT GTC GGG TTG TCC ATA GAG CCG CTC CTA TTG GGA AGC GGA GTG CAG TAT GAA AGC AGA GTT 2122
tetM  +++ ++T ++A ++G +++ +++ +++ +++ +++ ++C A+T ++T C+A ++T G++ +CA +++ ++T +C+ ++A ++G +++ +++ ++A +++ +++ ++G +++ TCG +++ 1522

TETO  SER LEU GLY TYR LEU ASN GLN SER PHE GLN ASN ALA VAL MET GLU GLY VAL LEU TYR GLY CYS GLU GLN GLY LEU TYR GLY TRP LYS VAL
tetO  TCA CTT GGA TAT TTA AAT CAA TCG TTC CAA AAT GCG GTT ATG GAG GGG GTT CTT TAT GGC TGC GAG CAG GGG CTG TAT GGA TGG AAA GTG 2212
tetM  ++T +++ +++ ++C +++ +++ +++ +++ ++T +++ +++ ++A +++ +++ +++ +++ A+A +GC +++ +++ ++T ++A ++A ++A T++ +++ +++ ++T +++ ++T 1612

TETO  THR ASP CYS LYS ILE CYS PHE GLU TYR GLY LEU TYR TYR SER PRO VAL SER THR PRO ALA ASP PHE ARG LEU LEU SER PRO ILE VAL LEU
tetO  ACA GAC TGT AAA ATC TGT TTT GAA TAT GGA TTG TAT TAT AGT CCT GTA AGT ACC CCC GCA GAC TTT CGG CTG CTT TCC CCT ATC GTA TTG 2302
tetM  ++G +++ +++ +++ +++ +++ +++ A+G +++ ++C ++A ++C +++ ++C +++ ++T +++ +++ ++A +++ ++T +++ +++ A++ +++ G+T +++ ++T +++ +++ 1702

TETO  GLU GLN ALA LEU LYS LYS ALA GLY THR GLU LEU LEU GLU PRO TYR LEU HIS PHE GLU ILE TYR ALA PRO GLN GLU TYR LEU SER ARG ALA
tetO  GAG CAG GCT TTA AAA AAA GCA GGG ACA GAA CTA TTA GAG CCA TAT CTC CAC TTT GAA ATT TAT GCA CCG CAG GAA TAT CTC TCA CGG GCG 2392
tetM  ++A ++A +TC +++ +++ +++ ++T ++A +++ +++ T+G +++ +++ +++ +++ ++T AGT +++ A++ +++ +++ ++G ++A +++ +++ +++ ++T +++ ++A ++A 1792

TETO  TYR HIS ASP ALA PRO ARG TYR CYS ALA ASP ILE VAL SER THR GLN ILE LYS ASN ASP GLU VAL ILE LEU LYS GLY GLU ILE PRO ALA ARG
tetO  TAT CAT GAT GCT CCA AGG TAT TGT GCA GAT ATT GTA AGT ACT CAG ATA AAG AAT GAC GAG GTC ATT CTG AAA GGA GAA ATC CCT GCT AGA 2482
tetM  ++C A+C +++ +++ ++T +AA +++ +++ ++G A+C ++C +++ CAC +++ ++A T+C ++A +++ A+T +++ +++ +++ ++T +CT +++ +++ +++ +++ +++ C+G 1882

TETO  CYS ILE GLN GLU TYR ARG ASN ASP LEU THR ASN PHE THR ASN GLY GLN GLY VAL CYS LEU THR GLU LEU LYS GLY TYR GLN PRO ALA ILE
tetO  TGT ATT CAA GAA TAC AGG AAC GAT TTA ACT AAT TTC ACA AAT GGG CAG GGA GTC TGC TTG ACA GAG TTA AAA GGA TAC CAG CCA GCT ATT 2572
tetM  +++ +++ +++ +++ +++ ++T C+T +GT +++ +++ +++ TTC ++T +++ +++ ++A +GT A+T ++T ++T ++A +++ +++ +++ +++ ++G +++ ++T GTT A++ +CC 1972

TETO  GLY LYS PHE ILE CYS GLN PRO ARG ARG PRO ASN SER ARG ILE ASP LYS VAL ARG HIS MET PHE HIS LYS LEU ALA ***
tetO  GGT AAA TTT ATT TGC CAA CCC CGC CGC CCG AAT AGC CGT ATA GAT AAG GTT CGG CAT ATG TTC CAC AAG TTA GCT TAA CAGCTTGCAA AAGT 2664
tetM  +++ G++ CC+ G++ +++ ++G +++ ++T ++T ++A +++ ++T ++G +++ +++ ++A ++A ++A T++ +++ +++ A+T ++A A++ A++ ++G TGTA++TT+T GTTG 2064

      CATATA AAATGAGATT TGAAAGGATT AGA 2693
      TTATAT ++++AT+G++ +CTTGTT+AA TA+ 2093
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Table 2

Codon usage in the *tetO* structural gene

Phe	TTT 15	Ser	TCT 10	Tyr	TAT 26	Cys	TGT 06
	TTC 06		TCC 04		TAC 03		TGC 05
Leu	TTA 18		TCA 09		***TAA 01		***TGA 00
	TTG 16		TCG 02		***TAG 00	Trp	TGG 04
Leu	CTT 11	Pro	CCT 09	His	CAT 09	Arg	CGT 06
	CTC 05		CCC 04		CAC 05		CGC 02
	CTA 06		CCA 08	Gln	CAA 11		CGA 00
	CTG 08		CCG 09		CAG 21		CGG 06
Ile	ATT 29	Thr	ACT 05	Asn	AAT 15	Ser	AGT 07
	ATC 09		ACC 04		AAC 10		AGC 06
	ATA 20		ACA 20	Lys	AAA 27	Arg	AGA 07
Met	ATG 15		ACG 07		AAG 11		AGG 07
Val	GTT 14	Ala	GCT 08	Asp	GAT 20	Gly	GGT 05
	GTC 06		GCC 03		GAC 09		GGC 06
	GTA 14		GCA 17	Glu	GAA 35		GGA 15
	GTG 08		GCG 04		GAG 18		GGG 14

quenced and the nucleotide sequence obtained by computer analysis is shown in Fig. 2. A single open reading frame of 1917 bp was identified by analysis of the termination and initiation codons in the three reading frames on each strand of DNA sequenced. The deduced protein which starts at the ATG codon at position 731 and extends to the termination codon TAA at position 2648 contains 639 amino acids and has a calculated molecular weight of 72 290. This value is closely related to that of 72 500 obtained for the TETM protein [8]. A putative ribosome binding site (RBS) sequence (TGATAGGAGG) is located 8 bp upstream from the translation initiation codon. A computer search for literal homology with the −35 (TTGACA) and −10 (TATAAT) consensus sequences revealed a canonical promoter at positions 421 and 444 (Fig. 2).

The *tetO* and *tetM* genes were compared by constructing optimal alignments between their nucleotide and amino acid sequences. The *tetO* gene exhibits 76% sequence identity with *tetM* (Fig. 2). Comparison of the sequences reveals that the nucleotide substitutions are scattered throughout, suggesting that the two genes have diverged from a common ancestor.

It has been recently reported [21] that the nucleotide sequence of the *aph*A-3 gene, also carried by pIP1433, and those from *Staphylococcus*, *Streptococcus* and *Pneumococcus* [22] were identical. This result was interpreted as the demonstration of the extension of the resistance gene pool of Gram-positive cocci to *Campylobacter* [9,21]. The similarity observed between *tetM* and *tetO* supports the claim that the tetracycline resistance gene of *Campylobacter* also originated in a Gram-positive bacterium. In fact, hybridization experiments using a probe specific for *tetO* (see below) indicated that this gene is present in various species of *Streptococcus* and in *Enterococcus* (Zilhao and Courvalin, manuscript in preparation). This observation is compatible with the hypothesis that presence of *tetO* in *Campylobacter* results from the acquisition of genetic material from Gram-positive bacteria under natural conditions.

The codon usage in *tetO* is presented in Table 2. There is a strong preference for AT rich codons among codons specifying the same amino acid. This non-random codon usage reflects the low GC content of the coding sequence (39%) and is similar to that in *tetM* [8] and in a chromosomal gene of *Enterococcus* (R. Wirth, unpublished result). Moreover, the RBS sequence of *tetO* is complementary to 8 (underlined) out of 10 bases of the 3′-OH terminus of the 16S rRNA (3′-OH UCUUUCCUCC-5′) of *Bacillus subtilis*, a feature which distinguishes genes from Gram-positive and Gram-negative bacteria [23]. These observations constitute further support for the notion of an origin in Gram-positive bacteria of the *tetO* gene.

The deduced amino acid sequences of *tetO* and *tetM* are 77% homologous. The product of *tetO* (TETO), as well as the TETM protein [8], is

Fig. 2. Nucleotide sequence of the DNA region containing the *tetO* gene of pIP1433 and comparison with *tetM* of Tn1545. Identity with the DNA sequence of *Campylobacter* is indicated by plus. Dashes represent gaps introduced to ensure optimal homology. The first base pair in the sequence of the fragment containing *tetO*, as presented, is defined as position 359. The presumed ribosome binding site (RBS), optimally spaced (8-bp) before the initiation codon, is boxed. The TTGACA and TATAAT consensus sequences are underlined. The deduced aminoacid sequence of the TETO protein is presented.

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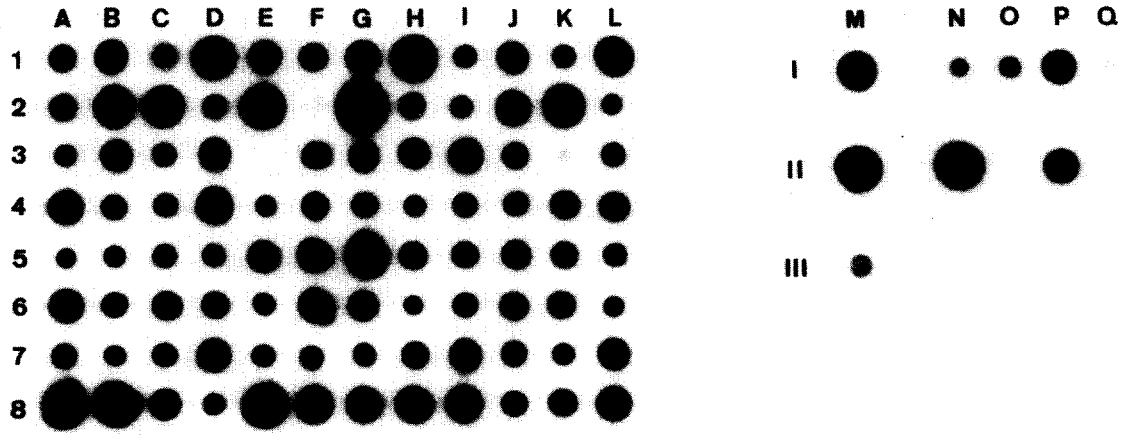

Fig. 3. Analysis of DNA by dot blot hybridization. Total DNA of *Campylobacter* strains was transferred to a nitrocellulose sheet and hybridized to the *tetO* probe labelled with ^{32}P in vitro (4×10^6 cpm per probe). Columns A to L and lane I (M to P) correspond to 100 Tc^R *Campylobacter* strains. 2-F and 3-E correspond to the two *C. jejuni* isolates which did not hybridize to the probe. II-M, DNA of the *tetO* probe; II-N, pAT121 (pUC8Ω*tetO*); II-O, pUC8; II-P, pIP1433 (*tetO*); II-Q, JH2-2::Tn1545 (*tetM*); III-M is a duplicate, with more DNA, of 3-K which hybridized weakly; III-N to Q susceptible control strains.

hydrophilic with an hydropathy value of −0.18 calculated according to Kyte and Doolittle [24]. The *tetM* determinant has been shown to mediate Tc^R at the level of protein synthesis [3] and it is likely that an analogous mechanism is involved in the Tc^R of *Campylobacter.*

4.3. Distribution of tetO among Campylobacter spp.

We determined the MIC's of Tc and minocycline against one hundred Tc^R *C. coli* and *C. jejuni* strains. In all isolates, resistance to tetracycline ($32 \leq MIC \leq 1024$ μg/ml) was associated with low level resistance to minocycline ($4 \leq MIC \leq 32$ μg/ml). Thus, the Tc^R determinant in *Campylobacter,* as *tetM* and *tetN* in streptococci [3], confers resistance to tetracycline but also to its lipophilic analogs. Tetracycline resistance in *Campylobacter* is known to be borne by plasmids able to conjugate to other *Campylobacter* spp. [5,6,25]. We have examined the transferability of Tc^R in twenty *Campylobacter* strains and all the strains were able to conjugate Tc^R to BM2560.

We screened the presence of *tetO* among *Campylobacter* strains using as a probe the intragenic 1458-bp *Hin*dIII-*Nde*I fragment of the insert in pAT121 (Fig. 1). Nucleotide sequences structurally related to *tetO* were detected in 98 strains under high stringency conditions (Fig. 3). Two Tc^R *C. jejuni* isolated from humans in France and Switzerland did not reveal homology with the probe, under the same hybridization conditions. None of the Tc susceptible strains hybridized with the probe (Fig. 3).

Certain strains of *Staphylococcus, Streptococcus* and *Enterococcus* contain more than one Tc^R determinant (R. Bismuth et al., manuscript in preparation). This observation combined with the fact that *tetO* may originate in Gram-positive cocci, prompted us to examine the possibility that nucleotide sequences structurally related to the other Tc^R determinants of Gram-positive bacteria were also present in *Campylobacter* spp. Specific probes consisting in fragments internal to the Tc^R structural genes of *Staphylococcus aureus* plasmid pT181 (tentatively called *tetK*), *Bacillus cereus* plasmid pBC16 (*tetL*) and of streptococcal conjugative shuttle transposon Tn1545 (*tetM*) were constructed (Table 1). Plasmid pMV120 from *S. agalactiae* was used as a probe to test the presence

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of *tetN*. We did not detect any hybridization between probes *tetK*, *tetL* and *tetN* and the *Campylobacter* strains including the two *C. jejuni* which did not hybridize with the *tetO* probe. However, because of homology between *tetO* and *tetM* (Fig. 2), we observed hybridization between the *tetM* probe and total DNA of *Campylobacter* strains harbouring *tetO* under low stringency but not under high stringency conditions (data not shown).

We did not search for Tc^R determinants of Gram-negative bacteria in *Campylobacter* since it has been reported that there is no homology between these genes and that of *Campylobacter* [5]. Our hybridization data indicate that neither *tetM* nor any of the other Tc^R determinants of Gram-positive bacteria is spread in *Campylobacter* spp. Gene *tetO* accounts for the vast majority (98%) of Tc^R resistant strains of *Campylobacter* but genetic heterogeneity for this character exists in this bacterial genus.

Based on plasmid stability, efficiency of transfer, restriction endonuclease analysis and Southern hybridizations, pIP1433 appears indigenous to *Campylobacter* [4,9,21]. Tetracycline resistance in *Campylobacter* results therefore from the acquisition of a gene rather than a replicon en bloc. We propose that, after an evolutionary process leading from a common ancestor to *tetM* and *tetO*, intergeneric transfer of DNA from a Gram-positive coccus to *Campylobacter* occurred. Subsequent dissemination of *tetO* in the latter genus most probably occurred via conjugation.

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